

Continuation Loci and the Origin of Local Domains

Pre-Topological Continuation-Neighborhood Structure under Standing-Bearing Persistence

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Abstract

Realized persistence inside a fixed admissible continuation regime cannot be reconstruction-null, and standing-bearing persistence under perturbation and restorative continuation constraints must factor through bounded localization classes. This paper derives the next physical role: local-domain organization. It asks whether bounded localization classes can remain unrelated atoms while still supporting lawful same-domain continuation among standing-bearing loci. They cannot.

The central result is a continuation-neighborhood induction theorem. Boundary compatibility is a same-domain reconstructibility condition on locus interfaces. Adjacency is the narrower condition in which such compatibility participates in standing-relevant continuation, comparison, constraint, or restoration. When bounded standing-localization classes participate in lawful boundary-compatible continuation, adjacency is induced as a continuation-relevant relation among loci. That induced adjacency generates pre-topological continuation-neighborhood assignments, standing-verdict separability, and local continuation domains.

The physical-role claim is that local physical domains first enter as continuation-neighborhood domains: organized neighborhoods of boundary-compatible standing-bearing loci before metric distance, topological open sets, manifolds, causal cones, field support, or local algebras are assumed. Later spacetime-local and field-local structures are downstream specializations of this local-domain role.

1 Introduction

1.1 Theorem claim and physical role

This article derives the local-domain role, not merely an abstract graph of admissibility. Paper 2 supplies bounded standing-localization. The present paper shows that bounded loci cannot support cross-locus standing-bearing continuation unless boundary-compatible adjacency and continuation-neighborhood domains are induced:

$$[B_D]_{\equiv_B} \wedge \text{BoundaryFix}_D \wedge \text{ContCompat}_D \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{Sep}_D, \text{LCD}_D).$$

The physical-role reading is: local physical domains originate as continuation-neighborhood domains. Topological neighborhoods, manifold charts, causal domains, field supports, and local algebra assignments are later specializations of this role.

The analysis continues the realized-persistence reconstruction sequence. The preceding non-vacuity result established a non-vacuity floor: a target typed as carrying realized persistence inside a fixed admissible continuation regime cannot remain null with respect to its own persistence-sensitive reconstruction structure [1]. In schematic form,

$$\text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D.$$

The standing-localization result established that non-vacuous standing-bearing persistence cannot remain merely global, diffuse, trace-only, hidden, or unbounded once standing-relevant perturbation and restorative continuation constraints are admitted [2]. In schematic form,

$$\text{NonVacRec}_D \wedge \text{RealPers}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv_B} \neq \emptyset.$$

That result is not metric locality, spacetime topology, or field support. It is bounded standing-localization: standing-bearing persistence factors through bounded continuation-sensitive loci over which perturbation burden, restoration, identity-memory, retained standing, and failure are jointly evaluated.

The next question is whether bounded standing-localization classes can remain mutually unrelated. A mere set of unrelated atoms cannot support lawful continuation, comparison, restoration, or standing-relevant constraint across loci. If two loci participate in the same standing-bearing continuation relation, their boundary traces must be jointly admissibly reconstructible. Boundary compatibility does not by itself amount to adjacency; when paired with standing-relevant continuation, it induces adjacency. Adjacency, in turn, induces a pre-topological continuation-neighborhood assignment and generates local continuation domains by admissible adjacency paths.

The result is intentionally weaker than a derivation of spacetime. It does not prove that the induced continuation-neighborhoods are open sets in a topological space, neighborhoods in a manifold, metric balls, causal diamonds, field supports, local algebras, or spatial regions. It establishes that bounded standing-localization classes cannot support lawful cross-locus continuation without a compatibility organization that plays the minimal continuation-neighborhood role: it determines which loci are adjacent for standing-relevant continuation, which loci remain separately evaluable absent admitted adjacency, and which families of loci belong to the same local continuation domain.

1.2 Main result

The controlling thesis is:

Continuation-neighborhood structure first enters not as topology, distance, or spatial extension, but as the admissible organization of boundary-compatible continuation among bounded loci whose standing-bearing persistence can be jointly continued, separated, or restored.

In compact form, the main implication is

$$[B_D]_{\equiv_B} \wedge \text{ContCompat}_D \wedge \text{BoundaryFix}_D \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{Sep}_D, \text{LCD}_D).$$

Here Adj_D is the continuation-relevant refinement of boundary compatibility, $\mathcal{N}_D$ is the induced pre-topological continuation-neighborhood assignment, Sep_D is standing-verdict separability, and LCD_D is the local continuation-domain assignment generated by admissible adjacency paths. The implication is not an empirical law, not a physical-distance theorem, and not a primitive topological theorem. It is a fixed-domain reconstruction constraint.

1.3 Scope

The argument establishes five claims.

- (i) **Bounded loci alone are insufficient for continuation-neighborhood structure.** A family of bounded standing-localization classes may exist without yet inducing adjacency or local domains.
- (ii) **Boundary-compatible continuation induces adjacency.** Boundary compatibility supplies the admissibility background; it becomes adjacency only when used by standing-relevant continuation, comparison, constraint, or restoration.
- (iii) **Adjacency induces continuation-neighborhood assignment.** For a locus L , the immediate continuation-neighborhood $\mathcal{N}_D(L)$ consists of L together with the loci adjacent to it in the continuation-relevant boundary-compatible sense.
- (iv) **Standing-verdict separability is induced by non-collapse across adjacency.** Distinct loci remain independently evaluable except through admitted adjacency paths. This is not Bell separability or tensor-product factorization.
- (v) **Local continuation domains are generated by admissible adjacency paths.** Adjacency need not be transitive. Domain membership is generated by finite or schematically finite chains preserving boundary compatibility, quotient behavior, standing retention, and fail-closed status.

The argument does not derive metric distance, an open-set topology, differentiable manifolds, Lorentzian structure, finite propagation speed, causal cones, local field algebras, particle support, gauge structure, vacuum structure, or empirical spacetime. It also does not assert local hidden variables, Bell-local factorization, classical separability, or the absence of entanglement. The result is pre-topological and pre-metric. It derives the minimal adjacency and continuation-neighborhood role that later topology, metricity, propagation, and field structure must refine if they are reconstructed downstream.

If primitive topology is assumed at the start, the result is trivial. If metric distance is imported, adjacency becomes coordinate nearness and the proof no longer explains the origin of continuation-neighborhood structure. If a hidden selector fixes adjacency from outside the fixed regime, the construction violates the AMetric boundary discipline inherited from AASC [6]. The construction therefore derives adjacency from boundary-compatible continuation relevance rather than presupposing it.

1.4 Relation to standard locality frameworks

Standard physical locality frameworks typically begin with a background space, spacetime region, causal relation, algebraic net, or operational scenario. Algebraic quantum field theory assigns algebras to spacetime regions and studies isotony, locality, covariance, and related constraints [12, 13]. Operational and information-theoretic reconstructions often begin with systems, measurements, probabilities, information-processing tasks, or compositional assumptions [15, 16, 17, 18]. Those approaches enter at a later reconstruction layer.

The present analysis targets an earlier question: before one has a topology, metric, field algebra, causal structure, or informational measurement interface, what minimal continuation-neighborhood-like structure must already be present if bounded standing-localization classes can continue, constrain, or restore standing-relevant consequence across one another? The required structure is continuation-relevant boundary-compatible adjacency. This is weaker than physical spacetime locality, but it is the structural role such locality must later refine.

2 Dependency order and terminology

The argument imports prior results only in their fixed-domain form. It does not rely on a hidden physical substrate or a canonical representative of a locus.

Imported result	Role in this manuscript
Minimal non-vacuity	Realized persistence requires non-trivial continuation-sensitive reconstruction; this supplies the non-vacuous floor [1].
Standing localization	Standing-bearing persistence under perturbation and restorative continuation constraints factors through bounded localization classes [2].
Continuation-locus induction	Non-degenerate continuation-bearing governance induces a locus-equivalence class rather than mere trace succession [3].
Boundary-trace fixation	Role anchors for continuation loci are fixed by declared, stable, same-domain boundary traces, not by representative selection [4].
Persistence-regime realization	Realized persistence is irreversible standing-bearing consequence retention rather than abstract trace classification [5].
AMetric boundary discipline	No metric, probability, selector, default representative, or canonicalization is available before it is admitted as fixed-domain structure [6].
Anchor-tensor-skin discipline	A diagrammatic or representational edge is not an admissible adjacency unless it carries anchor or tensor role load [7].
Concrete-domain closure	Standing-bearing persistence cannot be discharged into formal, trace-only, semantic-only, skin-only, global, diffuse, or bookkeeping closure while preserving the same target [8].
Fixed-domain exhaustion	Same-scope alternatives must preserve gate, target, quotient, transport, boundary typing, and fail-closed structure [9].

Throughout the manuscript, the word “neighborhood” means continuation-neighborhood unless a topological sense is explicitly indicated.

Term	Structural role
Admissibility	Lawful same-domain admission, rejection, continuation, redescription, and fail-closed status.
Standing	Persistence-bearing admissibility status retained through licensed continuation.
Continuation locus	Minimally preserved admissibility-relevant identity locus under lawful same-domain continuation.

Term	Structural role
Bounded standing-localization class	Bounded continuation-sensitive class through which standing-bearing persistence retains consequence under perturbation and restorative continuation constraints.
Boundary trace	Same-domain structural interface fixing role-compatible boundary behavior; not observation, measurement, coordinate edge, or semantic report.
Boundary compatibility	Joint reconstructibility of boundary traces under lawful continuation without identity reset, hidden selector import, representative choice, or domain drift; a compatibility condition, not adjacency by itself.
Adjacency	Continuation-relevant boundary compatibility among loci; a refinement of compatibility, not metric closeness, topology, spatial contact, graph edge by fiat, or field interaction.
Continuation-neighborhood	The immediate locus plus its admitted adjacent loci; pre-topological, not an open-set neighborhood.
Separability	Independent evaluability of standing-bearing continuation status absent admitted adjacency paths; not Bell separability.
Local continuation domain	Closure of a locus under admissible adjacency paths preserving boundary compatibility, quotient behavior, standing retention, and fail-closed status.

3 Fixed-domain preliminaries

The definitions below are intentionally weak. They do not assume metric, topological, causal, spatial, field-theoretic, probabilistic, graph-theoretic, or informational structure.

Definition 3.1 (Fixed admissible continuation regime). A fixed admissible continuation regime D is a comparison setting in which target identity, admissibility classification, standing-bearing continuation, boundary typing, quotient identity, transport behavior, lawful redescription, and fail-closed status are evaluated against the same invariant bundle. A proposal that changes the gate, target vocabulary, comparison relation, failure semantics, or evaluated fragment changes the regime.

Definition 3.2 (Continuation locus). A continuation locus L is the minimally preserved admissibility-relevant identity locus under lawful same-domain continuation. It is not a substance, object, spacetime point, material carrier, semantic denotatum, intentional target, or hidden bearer. It is the invariant role by which a regime distinguishes continuation of the same standing-relevant target from replacement, drift, relabeling, rejected trace, or scope change. Two loci are locus-equivalent, $L \equiv_{\text{Loc}} L'$, iff they preserve identical continuation eligibility, standing-bearing reuse, transport behavior, quotient behavior, boundary discipline, and fail-closed continuation structure under lawful same-domain redescription.

Definition 3.3 (Bounded standing-localization class). A bounded standing-localization class $B \in [B_D]_{\equiv_B}$ is a bounded continuation-sensitive class through which standing-bearing persistence retains consequence under perturbation and restorative continuation constraints. It carries the role profile required for retained consequence: boundary, identity-memory, perturbation burden, restorative continuation obligations, quotient stability, standing verdict, and fail-closed status.

Remark 3.4 (Boundedness). Boundedness is role-boundedness. It means that perturbation burden, restoration, retained identity, and failure verdict are evaluated over a delimited continuation-sensitive locus rather than over an unbounded global whole, an indeterminate aggregate, or an external selector. It does not mean metric boundedness, topological compactness, finite cardinality, spatial enclosure, physical extension, or coordinate diameter.

Definition 3.5 (Boundary trace). A boundary trace $\text{BT}_D(L)$ for a locus L is a declared same-domain continuation-fixation interface preserving boundary-relevant standing behavior under lawful same-domain redescription. It is admissible only when it is stable, same-domain typed, quotient-respecting, role-indexed, and fail-closed. It is not an observation, measurement outcome, semantic label, coordinate boundary, skin-level mark, or post-hoc selector.

Definition 3.6 (Boundary-compatible loci). Two bounded loci L_i, L_j are boundary-compatible, written

$$L_i \text{BC}_D L_j,$$

iff their boundary traces can be jointly admissibly reconstructed under lawful same-domain typing without identity reset, hidden selector import, representative selection, fail-open relabeling, or domain drift. Boundary compatibility is a boundary co-admissibility relation: it records that the two boundary profiles can be reconstructed together inside the fixed regime. It does not by itself assert continuation relevance, transfer, restoration, constraint, causal access, spatial nearness, or a graph edge.

Remark 3.7 (Compatibility is weaker than adjacency). Boundary compatibility and adjacency have different types. Boundary compatibility concerns joint reconstructibility of boundary traces. Adjacency concerns boundary-compatible *continuation relevance*. Thus

$$\text{Adj}_D \subseteq \text{BC}_D$$

in the role sense that every admitted adjacency must be boundary-compatible, while a boundary-compatible pair need not be adjacent unless standing-relevant continuation actually passes, compares, constrains, or restores across the compatible interface. This separation prevents adjacency from being introduced merely by renaming compatibility.

Definition 3.8 (Continuation compatibility). $\text{ContCompat}_D(L_i, L_j)$ holds when L_i and L_j participate in a lawful same-domain standing-relevant continuation relation whose retained consequence can pass, compare, constrain, or restore across their boundary interfaces while preserving quotient behavior, target identity, and fail-closed standing status.

Definition 3.9 (Adjacency). For bounded loci L_i, L_j , write

$$L_i \text{Adj}_D L_j$$

iff $L_i \text{BC}_D L_j$ and there exists a lawful same-domain continuation relation in which standing-relevant consequence can pass, compare, constrain, or restore across their boundary-compatible interface. Adjacency is therefore the continuation-relevant subrelation of boundary compatibility, not compatibility alone. It is not metric closeness, topological contact, spatial nearness, causal accessibility, field interaction, graph-theoretic primitive, or diagrammatic edge by fiat.

Definition 3.10 (Pre-topological continuation-neighborhood assignment). The pre-topological continuation-neighborhood of a locus L is

$$\mathcal{N}_D(L) = \{L\} \cup \{K : L \text{Adj}_D K\}.$$

The assignment $L \mapsto \mathcal{N}_D(L)$ is a continuation-neighborhood role assignment, not yet a topology. It need not satisfy open-set axioms, symmetry, transitivity, separation axioms, local finiteness, manifold compatibility, or metric representability.

Definition 3.11 (Standing-verdict separability). Two loci L_i, L_j are standing-verdict separable, written $\text{Sep}_D(L_i, L_j)$, when standing failure, perturbation burden, restoration obligation, or identity reset in L_i does not automatically determine the corresponding standing verdict in L_j except through admitted adjacency paths preserving boundary-compatible continuation. Operationally, separability concerns the independent evaluability of standing-bearing continuation status absent admitted adjacency structure. This is not Bell separability, tensor-product factorization, statistical independence, or ontic independence.

Definition 3.12 (Adjacency path and local continuation domain). An admissible adjacency path from L_0 to L_n is a finite or schematically finite chain

$$L_0 \text{Adj}_D L_1 \text{Adj}_D \cdots \text{Adj}_D L_n$$

whose intermediate boundary traces preserve standing-bearing quotient behavior, target identity, lawful continuation, restoration compatibility, and fail-closed status at every composition step. Arbitrary concatenation is not allowed: a chain is path-admissible only when each intermediate locus carries the quotient and boundary load required for same-domain standing-bearing continuation. The local continuation domain generated by L_0 is

$$\text{LCD}_D(L_0) = \{L_n : \exists \text{ an admissible adjacency path } L_0 \rightsquigarrow_{\text{Adj}_D}^* L_n\}.$$

This is not a spatial region. It is the continuation domain generated by admitted adjacency.

4 Atomized loci are insufficient

Bounded standing-localization is a necessary predecessor of continuation-neighborhood structure, but it is not the same as continuation-neighborhood structure. A mere family of bounded loci may be internally well formed while remaining mutually unrelated. Such a family has loci, but it has no adjacency, no cross-locus standing comparison, no local continuation-domain membership, and no separability relation except the trivial absence of any admitted relation.

Lemma 4.1 (Isolated-locus insufficiency). *Suppose $[B_D]_{\equiv_B} \neq \emptyset$, but for every pair of distinct bounded loci L_i, L_j there is no boundary-compatible continuation relation between them. Then D contains bounded standing-localization classes but does not yet contain a continuation-neighborhood structure.*

Proof. A bounded standing-localization class supplies a delimited locus for perturbation burden, restoration evaluation, retained standing, quotient behavior, and fail-closed status. It does not by itself specify whether another locus can compare, constrain, restore, or continue standing-relevant consequence with it. If no pair of distinct loci has boundary-compatible continuation, no adjacency relation is induced. Without adjacency there is no non-trivial continuation-neighborhood assignment beyond singleton self-inclusion, no non-trivial admissible adjacency path, and no local continuation domain containing more than its generator. The family of loci remains atomized. Hence bounded localization alone is insufficient for continuation-neighborhood structure. $\square$

Corollary 4.2 (Atomized localization is not locality structure). *A set or quotient family of bounded localization classes is not, by itself, a local continuation-domain structure. Continuation-neighborhood structure requires boundary-compatible continuation relevance among loci.*

Remark 4.3 (No overclaim from bounded localization). The standing-localization result derives bounded pre-metric localization. The present analysis does not rename those bounded loci as continuation-neighborhoods. It adds the cross-locus compatibility-and-use condition required for adjacency. This prevents the arc from overclaiming: non-vacuity is not localization, localization is not continuation-neighborhood structure, and continuation-neighborhood structure is not yet metric topology.

5 Boundary-compatible continuation relevance induces adjacency

The central point is that cross-locus continuation cannot be evaluated without boundary compatibility, while boundary compatibility alone is not yet adjacency. If standing-relevant consequence can pass, compare, constrain, or restore across loci, then the regime must preserve the boundary behavior by which such cross-locus participation remains same-domain and quotient-stable. Adjacency is the induced relation obtained when this boundary compatibility is used in a standing-relevant continuation role.

Proposition 5.1 (Boundary-compatible continuation relevance induces adjacency). *Let L_i, L_j be bounded standing-localization loci in a fixed admissible continuation regime D . If L_i and L_j participate in a lawful same-domain standing-relevant continuation relation, then their boundary traces must be jointly admissibly reconstructible, and that compatibility is activated as adjacency. Consequently,*

$$\text{ContRel}_D(L_i, L_j) \wedge \text{BoundaryFix}_D(L_i, L_j) \Rightarrow (L_i \text{BC}_D L_j \wedge L_i \text{Adj}_D L_j).$$

Boundary compatibility without the standing-relevant continuation conjunct does not imply adjacency.

Proof. A lawful cross-locus continuation relation purports to preserve standing-relevant consequence between L_i and L_j without changing the fixed domain. Such preservation cannot be evaluated if the boundary behavior of the two loci is unavailable, unstable, outcome-conditioned, representative-dependent, hidden, or cross-domain. Without a jointly reconstructible boundary interface, the putative relation cannot distinguish same-domain continuation from relabeling, identity reset, external transfer, default selection, or domain drift. A boundary-compatible interface is therefore required. The additional standing-relevant continuation relation activates that interface as adjacency. Since adjacency is defined as continuation-relevant boundary compatibility, the jointly reconstructible boundary-compatible continuation relation induces $L_i \text{Adj}_D L_j$, while boundary compatibility alone would only yield $L_i \text{BC}_D L_j$. $\square$

Corollary 5.2 (No adjacency without boundary work). *A diagram edge, syntactic successor relation, coordinate nearness claim, semantic association, probability weight, or informal similarity relation does not constitute adjacency unless it carries both boundary compatibility and standing-relevant continuation use inside the fixed admissible regime.*

Remark 5.3 (Adjacency is not imported). The order of construction is

$$\begin{aligned} \text{boundary-compatible continuation relevance} &\Rightarrow \text{adjacency} \\ &\Rightarrow \text{continuation-neighborhood assignment,} \end{aligned}$$

not

topological space $\Rightarrow$ adjacency.

This reversal is the main structural point.

6 No primitive topology or graph primitive

The term “continuation-neighborhood” is qualified because unqualified “neighborhood” naturally suggests topological neighborhoods, spatial surroundings, or metric balls. None of those meanings is used here. In topology, a neighborhood is ordinarily defined relative to open sets or an already available topology [14]. In algebraic quantum field theory, locality is ordinarily formulated over regions of spacetime with assigned algebras [12, 13]. The present analysis begins at the prior structural layer.

Criterion 6.1 (No topology import). A candidate continuation-neighborhood assignment $\mathcal{N}_D$ is admissible in this analysis only if it is induced by boundary-compatible continuation relevance among bounded standing-localization loci. It is inadmissible as a primitive import if its membership is fixed by any of the following without same-domain admissibility support:

- (a) coordinate nearness;
- (b) metric distance;
- (c) diagram placement;
- (d) syntactic succession;
- (e) semantic similarity;
- (f) probability weight;
- (g) hidden selector;
- (h) canonical representative choice;
- (i) topological open-set membership;
- (j) graph edge by fiat.

Remark 6.2 (Compatibility with later topology). The induced continuation-neighborhood assignment may later be represented by a topology, Alexandrov-style preorder, graph, simplicial complex, causal set, algebraic net, or metric structure if additional reconstruction constraints justify that representation. The present result does not deny those later representations. It states only that their admissible role begins at boundary-compatible continuation relevance among bounded loci.

7 Continuation-neighborhood induction

The main theorem is not that all bounded loci are adjacent. Nor is it that adjacency is symmetric, transitive, metric, topological, causal, or field-like. It states that whenever bounded standing-localization classes support lawful boundary-compatible continuation relevance among loci, a pre-topological continuation-neighborhood assignment is induced.

Theorem 7.1 (Continuation-neighborhood induction). *Let D be a fixed admissible realized persistence regime, and suppose:*

- (i) $[B_D]_{\equiv_B} \neq \emptyset$: *bounded standing-localization classes exist;*
- (ii) ContCompat_D : *lawful standing-relevant continuation among some bounded loci is in force;*
- (iii) BoundaryFix_D : *boundary traces for the participating loci are fixed in the same-domain admissible sense.*

Then D induces a non-trivial pre-topological continuation-neighborhood structure:

$$[B_D]_{\equiv_B} \wedge \text{ContCompat}_D \wedge \text{BoundaryFix}_D \Rightarrow \exists \mathcal{N}_D.$$

The induced assignment is

$$\mathcal{N}_D(L) = \{L\} \cup \{K : L \text{Adj}_D K\}$$

for each locus L participating in the boundary-compatible continuation relation.

Proof. By the standing-localization result, $[B_D]_{\equiv_B} \neq \emptyset$ supplies bounded continuation-sensitive loci capable of retaining standing-bearing consequence under perturbation and restorative continuation constraints. By ContCompat_D , at least one standing-relevant relation among such loci purports to preserve same-domain continuation, comparison, constraint, or restoration across locus boundaries. By BoundaryFix_D , the boundary behavior of the participating loci is declared, stable, quotient-respecting, and same-domain reconstructible. The continuation-relevance proposition then yields adjacency for each participating pair. Once adjacency is induced, the continuation-neighborhood assignment $L \mapsto \{L\} \cup \{K : L \text{Adj}_D K\}$ is defined. Since at least one cross-locus boundary-compatible continuation-relevance relation is in force, the assignment is non-trivial on the participating fragment. It is pre-topological because no open-set axioms, metric, manifold, causal, or field structure is assumed or derived. Hence a pre-topological continuation-neighborhood structure is induced. $\square$

Corollary 7.2 (Continuation-neighborhoods are quotient-level roles). *The induced continuation-neighborhoods are invariant under lawful same-domain redescription of the participating loci. They are role assignments over continuation-locus equivalence classes, not privileged skin-level representatives.*

Proof. Adjacency depends on boundary-compatible continuation relevance. Boundary compatibility and continuation relevance are defined modulo lawful same-domain redescription, quotient behavior, and fail-closed status. Hence replacing a locus by an equivalent representative preserves the induced adjacency and continuation-neighborhood assignment. $\square$

8 Standing-verdict separability

Continuation-neighborhood structure has a negative side as well as a positive side. If every standing verdict automatically globalized to every locus, adjacency would collapse into global standing and separability would vanish. Conversely, if no admitted relation can carry standing-relevant consequence across loci, the loci remain atomized. The useful structure lies between these degeneracies: standing verdicts are independently evaluable except through admitted adjacency paths. Operationally, separability concerns the independent evaluability of standing-bearing continuation status absent admitted adjacency structure; it is not isolation, ontic independence, or factorization of all underlying compatibility.

Lemma 8.1 (Failure non-propagation without adjacency). *Let L_i, L_j be bounded standing-localization loci. If there is no admitted adjacency path from L_i to L_j , then standing failure, perturbation burden, restoration obligation, or identity reset in L_i cannot automatically determine the corresponding standing verdict in L_j without importing global collapse, hidden transfer, or domain drift.*

Proof. An automatic cross-locus standing verdict requires a route by which standing-relevant consequence is carried from one locus to the other. In this framework such a route must be an admitted boundary-compatible continuation relation or a finite composition of such relations. If no adjacency path exists, then the purported automatic determination has no same-domain transport channel. It must either globalize all standing verdicts, import a hidden transfer relation, rely on a default selector, reset identity, or change the evaluated domain. Each route violates the fixed-domain conditions. Therefore the verdict cannot automatically propagate. $\square$

Theorem 8.2 (Standing-verdict separability). *In a fixed admissible realized persistence regime with induced adjacency Adj_D , bounded loci are standing-verdict separable whenever their standing verdicts remain independently evaluable except through admitted adjacency paths. In particular, if no admitted adjacency path connects L_i and L_j , then*

$$\text{Sep}_D(L_i, L_j)$$

with respect to automatic standing collapse.

Proof. The separability relation is defined by independent evaluability of standing failure, perturbation burden, restoration obligation, and identity status except through admitted adjacency paths. The preceding lemma establishes that absent such a path, automatic verdict transfer would require an illicit global, hidden, default, reset, or cross-domain route. Therefore the two loci are separable in the standing-verdict sense. $\square$

Remark 8.3 (Not Bell separability). Standing-verdict separability is not Bell separability, local hidden-variable factorization, tensor-product independence, statistical independence, or classical separability. It is a fixed-domain claim about automatic standing verdict collapse. The result is compatible with nonlocal tensor-level compatibility and with quantum entanglement in the ordinary Bell-classification sense [11, 10].

9 Local continuation domains

Adjacency need not be transitive. If $L_1 \text{Adj}_D L_2$ and $L_2 \text{Adj}_D L_3$, it does not follow that $L_1 \text{Adj}_D L_3$. What follows, under admissible composition, is that all three loci may belong to the same local continuation domain. This distinction prevents the theory from collapsing adjacency into all-to-all relatedness.

Definition 9.1 (Admissible composition of adjacency). Two adjacency links

$$L_1 \text{Adj}_D L_2, \quad L_2 \text{Adj}_D L_3$$

compose admissibly when the intermediate locus L_2 preserves standing-relevant boundary behavior, quotient status, target identity, restoration compatibility, and fail-closed treatment across both links. The composite is an adjacency path, not necessarily a direct adjacency.

Proposition 9.2 (Adjacency composition). *If $L_1 \text{Adj}_D L_2$ and $L_2 \text{Adj}_D L_3$ compose admissibly, then L_1, L_2, L_3 belong to the same local continuation domain generated by L_1 . Direct adjacency between L_1 and L_3 is not implied.*

Proof. The two links give an admissible adjacency path $L_1 \text{Adj}_D L_2 \text{Adj}_D L_3$. By definition of $\text{LCD}_D(L_1)$, every terminal locus of an admissible adjacency path from L_1 belongs to $\text{LCD}_D(L_1)$. Thus $L_3 \in \text{LCD}_D(L_1)$, and L_1, L_2, L_3 belong to the same generated domain. Since adjacency is defined by direct boundary-compatible continuation relevance, and no direct boundary-compatible relation between L_1 and L_3 has been given, direct adjacency is not inferred. $\square$

Theorem 9.3 (Local continuation-domain induction). *If a bounded standing-localization locus L_0 admits non-trivial admissible adjacency paths, then L_0 induces a local continuation domain*

$$\text{LCD}_D(L_0) = \{L_n : L_0 \rightsquigarrow_{\text{Adj}_D}^* L_n\}$$

closed under admissible finite adjacency composition.

Proof. The domain is defined as the set of loci reachable from L_0 by admissible adjacency paths. If $K \in \text{LCD}_D(L_0)$ and $K \text{Adj}_D M$ composes admissibly with the path from L_0 to K , then appending the link yields an admissible path from L_0 to M . Therefore $M \in \text{LCD}_D(L_0)$. The domain is closed under admissible finite adjacency composition by construction. It is a local continuation domain because membership is generated only by boundary-compatible standing-bearing continuation relevance, not by arbitrary set inclusion, primitive topology, metric proximity, or diagrammatic edge placement. $\square$

Remark 9.4 (Schematic finiteness). The phrase finite or schematically finite is used to avoid importing temporal discreteness or a countable underlying space. It means that the continuation step can be represented by a bounded admissible composition schema on the evaluated fragment. Later work may impose stronger finiteness, continuity, completion, or limiting conditions if metric or topological reconstruction requires them.

10 No default continuation-neighborhoods

The AMetric boundary discipline is crucial here. A continuation-neighborhood cannot be chosen by saying that two loci are close, similar, co-present, graph-adjacent, or co-indexed unless those relations are admitted as fixed-domain continuation structure. Without boundary-compatible continuation relevance, such choices are representative selection rather than continuation-neighborhood induction.

Theorem 10.1 (No default continuation-neighborhood selection). *A pre-topological continuation-neighborhood assignment $\mathcal{N}_D$ for bounded standing-localization loci cannot be selected by representative labels, diagram placement, coordinate proximity, semantic similarity, probability, observer report, arbitrary graph edge, hidden metric, or canonical representative choice unless that structure is admitted as same-domain boundary-compatible continuation relevance. In symbols,*

$$\text{Default}(\mathcal{N}_D) \wedge \neg \text{BoundaryFix}_D \Rightarrow \neg \text{Adm}_D(\mathcal{N}_D).$$

Proof. A default continuation-neighborhood selection determines which loci count as adjacent. If the selection is not fixed by same-domain boundary-compatible continuation relevance, then it supplies adjacency verdicts from outside the admissible continuation regime. Such a verdict

either imports an unlicensed selector, privileges a skin-level representative, relies on a hidden metric or diagram, or changes the evaluated domain. Each option violates the fixed-domain and AMetric boundary conditions. Therefore the default selection is not an admissible continuation-neighborhood assignment. $\square$

Corollary 10.2 (Skin-only adjacency failure). *A drawn edge, coordinate label, notation convention, or diagrammatic proximity does not carry adjacency unless it preserves the anchor or tensor role required for boundary-compatible standing-bearing continuation relevance.*

11 Global adjacency collapse and isolated-locality failure

Two trivializations must be blocked. One makes every locus adjacent to every other locus. The other leaves every locus isolated. Neither gives the target continuation-neighborhood structure unless supported by boundary-compatible continuation relevance.

Lemma 11.1 (Universal adjacency collapse). *If every bounded locus is declared adjacent to every other bounded locus without boundary-compatible continuation-relevance conditions, adjacency becomes indistinguishable from global standing and loses standing-verdict separability.*

Proof. If all loci are adjacent by declaration alone, then every standing verdict can be treated as immediately relevant to every other locus without any boundary-compatible relation carrying that relevance. The declared relation no longer distinguishes admitted cross-locus continuation from global verdict collapse. Since separability requires independent standing-verdict evaluability except through admitted adjacency paths, universal adjacency by fiat erases the distinction between direct compatibility and global standing. Unless every pair genuinely possesses boundary-compatible continuation-relevance structure, universal adjacency is inadmissible. $\square$

Lemma 11.2 (Isolated loci do not form a continuation domain). *A family of bounded loci with no admitted adjacency relations is merely a family of localization classes. It does not form a non-trivial local continuation domain.*

Proof. With no admitted adjacency relations, the only generated paths are singleton paths. Each $\text{LCD}_D(L)$ is therefore either the singleton $\{L\}$ or undefined on non-participating fragments, depending on whether self-inclusion is retained as a notational convention. No cross-locus continuation domain is induced. Hence isolated loci do not form a non-trivial local continuation domain. $\square$

12 Finite witness model

A finite model makes the structural distinction visible. The finite witness graph is illustrative only; it does not define the underlying continuation-neighborhood structure. It is useful because, when its edges are interpreted correctly, it directly instantiates the theorem architecture: bounded loci, boundary traces, adjacency, continuation-neighborhoods, separability, local continuation domains, and the excluded collapse cases.

Let

$$G_D = (V, E)$$

where vertices V are bounded standing-localization classes and edges E are adjacency relations, namely boundary-compatible pairs that also carry standing-relevant continuation use. For $v \in V$, let $\text{BT}(v)$ be its boundary trace, $s(v)$ its standing verdict, $p(v)$ its perturbation burden, and

$r(v)$ its restorative continuation profile. An edge $(v, w) \in E$ is admitted only when $\text{BT}(v)$ and $\text{BT}(w)$ are jointly reconstructible and the standing-relevant continuation relation between them preserves quotient behavior, target identity, restoration compatibility, and fail-closed status. Thus E represents Adj_D , not all of BC_D .

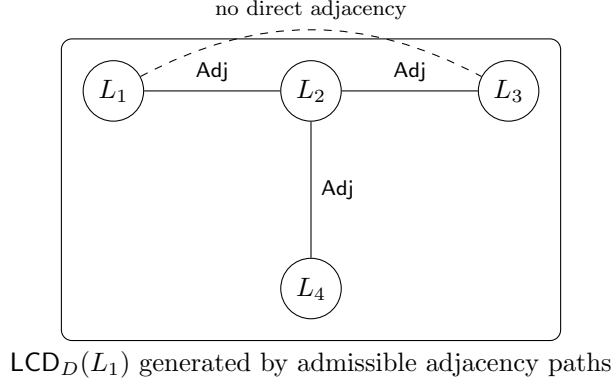


Figure 1: A finite witness model. Edges are adjacency relations: boundary-compatible pairs used by standing-relevant continuation, not metric distances or diagrammatic adjacency. The path $L_1 \text{Adj}_D L_2 \text{Adj}_D L_3$ places L_3 in $\text{LCD}_D(L_1)$ without implying direct adjacency $L_1 \text{Adj}_D L_3$.

The same vertices can represent three distinct quotient outcomes:

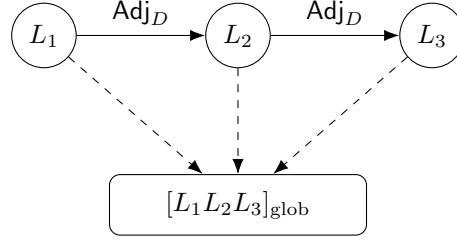
$$\begin{array}{ll}
 E = \emptyset & \text{isolated atoms,} \\
 E = V \times V & \text{global-collapse candidate,} \\
 E = \{(v, w) : v \text{Adj}_D w\} & \text{admitted continuation-neighborhood structure.}
 \end{array}$$

Only the last quotient form uses standing-relevant boundary-compatible continuation as the edge condition. The first has no non-trivial continuation-neighborhood structure; the second erases separability unless every pair really carries such continuation relevance.

Four cases capture the finite model's basic behavior.

- (i) **No edges.** Bounded loci exist, but no non-trivial continuation-neighborhood structure is induced. Each locus is isolated.
- (ii) **Complete graph by fiat.** All loci are declared adjacent without continuation-relevant boundary compatibility. This collapses adjacency into global standing and destroys separability.
- (iii) **Hidden edges.** Some edges are available only by unannounced selector, hidden metric, or representative choice. Such edges violate same-domain boundary fixation.
- (iv) **Admitted edges.** Edges are admitted exactly when boundary traces are jointly reconstructible and carry standing-relevant continuation. This yields the pre-topological continuation-neighborhood assignment $\mathcal{N}_D(v)$, standing-verdict separability across disconnected components, and local continuation domains as connected components under admissible adjacency paths.

Proposition 12.1 (Witness-model realization). *The finite graph model realizes the continuation-neighborhood induction theorem exactly when its edge set is the set of continuation-relevant boundary-compatible adjacency relations among bounded standing-localization vertices.*



quotient-collapse witness: declaring all loci adjacent by fiat replaces admissible path structure with a single global standing class and eliminates standing-verdict separability

Figure 2: A quotient-collapse witness. Admissible paths preserve intermediate boundary work; all-to-all or global quotienting erases the distinctions required for continuation-neighborhood structure.

Proof. If the vertices are bounded standing-localization classes and the edges are continuation-relevant boundary-compatible adjacency relations, then each edge instantiates Adj_D . The edge-induced continuation-neighborhood representation gives $\mathcal{N}_D(v)$. Admissible paths generate $\text{LCD}_D(v)$. Disconnected components give standing-verdict separability with respect to automatic verdict collapse. The no-edge, complete-fiat, and hidden-edge cases instantiate the isolated-locus insufficiency, universal adjacency collapse, and no-default-continuation-neighborhood results respectively. Hence the model realizes the theorem architecture rather than merely illustrating it. $\square$

13 Relation to spacetime locality

The present result is a predecessor of spacetime locality, not spacetime locality itself. It derives the role structure that any later spacetime-local interpretation must preserve:

- (i) bounded standing-localization loci;
- (ii) continuation-relevant boundary-compatible adjacency among loci;
- (iii) pre-topological continuation-neighborhood assignment;
- (iv) standing-verdict separability except through admitted adjacency paths;
- (v) local continuation domains generated by admissible adjacency composition.

It does not derive metric distance, spatial regions, topological open sets, manifold charts, causal cones, finite signal speed, Lorentz invariance, field support, or local algebra assignments. Those structures require additional constraints. The value of the theorem is that it identifies the minimal continuation-neighborhood role those later structures must refine rather than assume.

Remark 13.1 (Why “pre-topological” is the correct target). A topology supplies a family of open sets satisfying closure properties. A metric supplies distances that induce a topology. A causal structure supplies order-like constraints. None of those primitives is supplied here. It supplies adjacency as a boundary-compatible continuation-relevance relation and local domains as adjacency-path closures. That is weaker than topology, but sufficient to define the next reconstruction question: one can ask what stability, overlap, separation, and composition constraints are required for these local domains to admit topological or metric representation.

14 Relation to quantum and Bell locality

The separability result should not be read as a return to local hidden variables. Bell locality concerns probabilistic factorization conditions over hidden variables and measurement settings [11]. The present separability relation concerns standing-verdict propagation across bounded continuation loci. It says that failure, restoration, and standing status do not automatically globalize absent admitted adjacency structure. It does not assert that quantum correlations are locally factorizable, that entanglement is absent, or that tensor-level compatibility must decompose into independent local factors.

This distinction is important for downstream physics. Bounded standing-localization may coexist with boundary-level or tensor-level compatibility structures that are not Bell-local in the usual sense. The result does not settle those questions. It only prevents a different collapse: the collapse of all standing verdicts into global, unbounded, hidden, or default adjacency.

15 Continuation-neighborhood route exhaustion

The route-exhaustion result can be stated in compressed form. Its point is not to multiply exclusions, but to show that the admissible option has a unique role: a candidate continuation-neighborhood structure must be generated by continuation-relevant boundary compatibility among bounded loci. Failures fall into broad classes.

Candidate class	Status
Admitted adjacency	Continuation-relevant boundary-compatible route; induces Adj_D , $\mathcal{N}_D$, Sep_D , and LCD_D .
Isolated-locus failure	Bounded loci exist, but no non-trivial continuation-neighborhood structure is induced.
Global-adjacency collapse	All-to-all adjacency by fiat collapses adjacency into global standing unless every pair has genuine continuation-relevant boundary compatibility.
Hidden or default adjacency	Selectors, hidden metrics, representative choices, diagram edges, trace-only succession, or skin-only presentation import membership without fixed-domain boundary work.
Identity or domain failure	Cross-domain drift or identity reset changes target, gate, quotient, failure semantics, lineage, or evaluated fragment instead of preserving same-domain standing-bearing continuation.
Primitive metric/topology import	Assumes the target structure rather than deriving the pre-topological continuation-neighborhood role.

Theorem 15.1 (Continuation-neighborhood route exhaustion). *For bounded standing-localization classes inside a fixed admissible realized persistence regime, every same-scope candidate for continuation-neighborhood structure either induces continuation-relevant boundary-compatible adjacency or falls into isolated-locus failure, global-adjacency collapse, hidden/default adjacency, identity/domain failure, or primitive metric/topology import. Only continuation-relevant boundary-compatible adjacency supplies the pre-topological continuation-neighborhood role required for standing-bearing continuation among loci.*

Proof. Let $\mathcal{N}_D$ be a candidate continuation-neighborhood assignment. If its membership is induced by standing-relevant same-domain continuation across jointly reconstructible boundary traces, it is the admitted route and yields Adj_D . If no cross-locus relation is admitted, the loci remain isolated and no non-trivial continuation-neighborhood structure is present. If all loci are declared adjacent without boundary work, the relation globalizes and standing-verdict separability collapses. If membership is fixed by a selector, hidden metric, canonical representative, diagrammatic edge, probability weight, semantic association, trace-only succession, or skin-only presentation, then the assignment is hidden/default rather than fixed-domain continuation structure. If the route changes target, gate, quotient, transport, failure semantics, lineage, or evaluated fragment, or if it preserves apparent adjacency by replacing the lineage, it is identity/domain failure rather than same-domain continuation. If it begins with metric or topological neighborhoods, it assumes the later representational structure instead of deriving the pre-topological continuation-neighborhood role. No additional same-scope category remains: any proposed alternative either preserves continuation-relevant boundary compatibility or fails one of the fixed-domain conditions just listed. Therefore the continuation-relevant boundary-compatible route is exhaustive for the target result. $\square$

16 Normal form

The results can be summarized in a single normal form:

$$\boxed{[B_D]_{\equiv_B} \wedge \text{BoundaryFix}_D \wedge \text{ContCompat}_D \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{Sep}_D, \text{LCD}_D)}$$

where:

$$L_i \text{Adj}_D L_j \iff L_i \text{BC}_D L_j \text{ and a standing-relevant boundary-compatible continuation use exists,}$$

$$\mathcal{N}_D(L) = \{L\} \cup \{K : L \text{Adj}_D K\},$$

$$\text{Sep}_D(L_i, L_j) \text{ means no automatic standing-verdict collapse except through admitted adjacency paths,}$$

$$\text{LCD}_D(L) = \{K : L \rightsquigarrow_{\text{Adj}_D}^* K\}.$$

The second conjunct in the first line is essential: boundary compatibility alone is not adjacency. It becomes adjacency only when standing-relevant consequence can pass, compare, constrain, or restore across the relevant boundary interface.

All four objects are pre-topological and fixed-domain. The normal form does not assume distance, open sets, manifolds, fields, causal cones, physical regions, graph primitives, or local hidden variables.

17 Conclusion

Paper 1 established that realized persistence cannot be reconstruction-null. Paper 2 established that standing-bearing persistence under perturbation and lawful restoration must localize through bounded continuation-sensitive loci. The present paper derives the next physical role: local-domain organization.

Boundary compatibility is the admissible reconstructibility of locus interfaces. Adjacency is the narrower case in which that compatibility participates in standing-relevant continuation, comparison, constraint, or restoration. Under this distinction, bounded classes cannot support standing-bearing continuation relations among one another without continuation-relevant boundary-compatible adjacency. Such adjacency induces pre-topological continuation-neighborhood assignments, standing-verdict separability, and local continuation domains generated by admissible adjacency paths.

The physical role of the result is positive. The paper does not merely withhold topology, metric geometry, manifolds, causal cones, finite signal speed, field support, or local algebra assignments. It derives the prior role those structures later occupy: local continuation domains in which standing-bearing consequences can be separated, connected, compared, restored, and transported without hidden selection or global collapse.

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